

Chapter 1 - Hibernate Fundamentals

1. What is Hibernate?

Hibernate is an Object Relational Mapping (ORM) framework for Java. It maps Java objects to relational database tables and minimizes JDBC boilerplate code.

2. Why Hibernate?

- Eliminates repetitive JDBC code
- Automatically maps Java classes to database tables
- Supports transactions
- Provides caching (Level 1 and Level 2)
- Database independent
- Supports HQL, Criteria API and Native SQL
- Integrates with Spring MVC and Spring Boot

3. ORM (Object Relational Mapping)

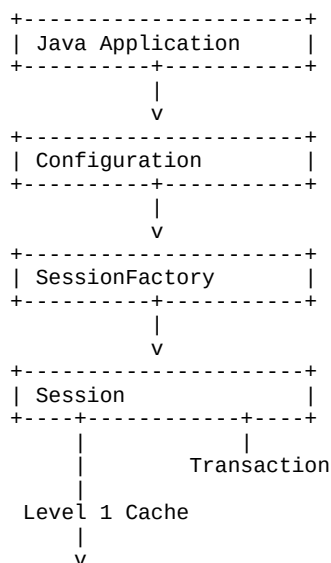
ORM maps Java objects to database tables. An object instance corresponds to a table row, while object fields correspond to table columns.

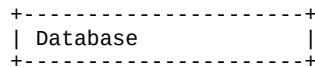
ORM Example

Java Object	Database Table
Employee	EMPLOYEE
-----	-----
id = 101 ----->	ID = 101
name = Siva ----->	NAME = Siva
salary=50000 ----->	SALARY = 50000

4. Hibernate Architecture

Main components: Configuration, SessionFactory, Session, Transaction and Database.





5. SessionFactory

SessionFactory is heavyweight, thread-safe, created once during application startup and reused to create Session objects.

6. Session

Session is lightweight, not thread-safe, represents a persistence context and contains the Level 1 Cache. It is typically created per request or transaction.

7. Transaction

Transactions ensure ACID properties. Common methods are beginTransaction(), commit() and rollback().

8. Basic CRUD Methods

save() - Insert
get() - Fetch by primary key
update() - Update detached entity
merge() - Merge detached state
remove()/delete() - Delete entity

9. Interview Points

- Hibernate is an ORM framework.
- JPA is a specification; Hibernate is an implementation.
- One SessionFactory per database.
- One Level 1 Cache per Session.
- Always use transactions for database modifications.